Aerial–Ground Mesh Networking System for Wilderness Search and Rescue

Instituto Superior Técnico
ElectroCap (PIC1)

icquackson.github.io/PIC-Website

Team (students & advisors)

Students	
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Problem definition

- Terrain and vegetation split ground teams into disconnected mesh partitions.
- Partitioned teams lose route continuity, voice coordination, and command visibility.
- Aerial elevation is required to restore connectivity when ground placement cannot bridge obstacles.
- Pre-planning goal: recommend initial UAV hover positions and minimum UAV count to cover/connect the target area (terrain/vegetation aware).
- Runtime goal: when partitions occur, UAV relay(s) autonomously reposition to restore connectivity without relying on operator connectivity.

Engineering objective

Implement UAV-assisted topology repair in a closed offline tactical mesh, including (i) pre-planning for initial relay placement and UAV count, and (ii) autonomous runtime self-repair that does not depend on operator connectivity.

System engineering scope

Novelty is end-to-end implementation and demonstration (UAV-assisted repair, autonomous workflow, acceptance evidence), not inventing a new routing protocol.

Solution beneficiaries

Direct beneficiaries	Final beneficiaries
▶ Firefighters ▶ Police/SAR units ▶ Field teams ▶ Command post operators	▶ Missing persons and casualties ▶ Responder safety during operations

Technological solution

- Ground mesh nodes + UAV relay node for topology repair.
- Open networking stack (Wi-Fi AP + BATMAN-adv) with local PTT/map/monitoring services.
- Autonomous UAV relay positioning: pre-planned initial hover points + runtime self-repair using live mesh telemetry (no operator dependency).
- Pre-planning tool estimates initial UAV hover positions and recommended number of UAV relays for a given area (engineering-grade model).
- Local services: PTT voice, map/position view, monitoring dashboard.
- Validated platform: hovering multirotor UAV carrying a self-powered relay payload (no flight-controller integration).
- Closed offline tactical mesh with no external infrastructure dependency.

Design principle

Simple, robust, repeatable deployment with deterministic autonomous repair behavior and measurable acceptance tests.

Networking & software stack

Networking stack

- User access: Wi-Fi AP on Team nodes (SSID + WPA2/WPA3).
- Mesh backbone: BATMAN-adv (L2) over 2.4 GHz Wi-Fi.
- Addressing: single L2 domain with per-node roles; fixed service IP on Command.
- Service discovery: mDNS/DNS (offline).
- Traffic handling: simple QoS (prioritize voice) using ‘tc‘/‘iptables‘ on the local mesh.

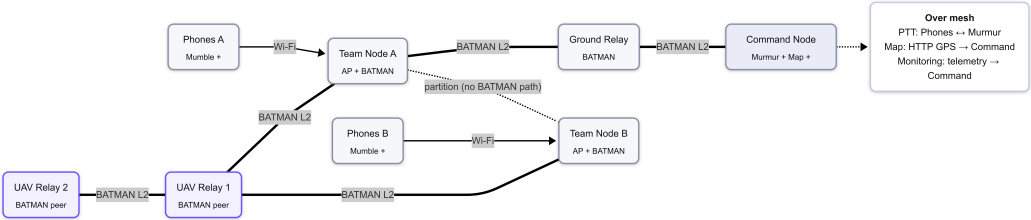
Software/services

- PTT voice: Murmur (server) on Command + Mumble client on phones.
- Map/position: local web dashboard (Leaflet or equivalent) with GPS upload endpoint.
- Monitoring/logging: node/link status, BATMAN originators, RSSI/SNR, uptime, battery logs.
- Automation: reproducible configs (OpenWrt UCI scripts / Ansible / shell) and one-command bring-up.
- Data capture: structured logs exported for the test report.

Why this stack

Simple, debuggable, repeatable offline implementation.

System architecture



Component integration

Node role	Network functions	Services & agents
Command	BATMAN-adv peer, fixed service IP, route/health aggregation	Murmur server, web map/dashboard, log collector, Autonomy Engine, admin UI
Team node	Wi-Fi AP for users, BATMAN-adv, QoS rules for voice	Telemetry Agent (GPS + link metrics), local AP management scripts
Ground relay	BATMAN-adv forwarding bridge, no user AP	Telemetry Agent, link/uptime telemetry exporter
UAV relay	BATMAN-adv relay bridge, fast boot-to-mesh procedure	Telemetry Agent, boot checklist for hovering multirotor payload

All services are local; closed offline network.

Architecture addendum: autonomy + telemetry blocks

- **Telemetry Agent (all nodes):** publishes GPS position + RSSI/SNR/TQ + node health over the local mesh.
- **Autonomy Engine (Command and/or UAV payload):**
  - Consumes telemetry and connectivity state to detect partitions.
  - Computes UAV move direction and altitude band to maximize link margin and reconnect partitions.
  - Can run distributed (on UAV payload) so repair works even if Command/operator links are unavailable.
- **Flight control boundary (v1):** autonomy implemented as waypoint nudges / position holds; no deep flight-controller integration required.

Hardware and telemetry

Telemetry signals collected

- RSSI/SNR per link and per route segment.
- BATMAN originators and next-hop changes over time.
- Packet loss/latency probes between command and team nodes.
- Battery runtime traces (voltage/current/runtime).
- GPS tracks from field teams and key relay positions.
- Reconvergence timestamps (partition detect, UAV join, route restore, PTT restore).

Data products for reports

- Coverage maps (RSSI/SNR vs location).
- Route-stability timelines and outage/recovery event tables.
- KPI plots for reconvergence, latency, and packet delivery.

Prototype hardware stack

Ground node (each)

- OpenWrt-capable router/SBC with 2.4 GHz Wi-Fi.
- External antenna: 2.4 GHz omni, vertical polarization.
- Connector chain: SMA/RP-SMA adapters + pigtails as needed.
- Mounting: small mast/clamp/tripod/zip ties.
- Power: USB power bank ($\geq$ 20,000 mAh) + short USB cable.
- Instrumentation: inline USB power meter or measured current-draw procedure.
- GPS source: phone GPS upload.
- Optional altitude source: barometer/altimeter.

UAV payload

- Validated platform: hovering multirotor UAV carrying a self-powered relay payload (no flight-controller integration).
- Mesh node + short/lightweight 2.4 GHz antenna.
- Self-powered battery pack sized for $\geq$ 30 min.
- Payload power measurement method: inline voltage/current logger during mission tests.
- Mounting bracket + vibration isolation + safe antenna clearance.
- Total payload mass target $<$ 500 g.

Node count

Command (1), Team nodes (2), Ground relay (1), UAV relay (1)

Operational constraints (explicit)

- Validated UAV platform: hovering multirotor, self-powered payload, no FC integration.
- Payload mass < 500 g (including mount + power).
- Boot-to-mesh ≤ 60 s from power-on.
- UAV mission window ≥ 30 min (battery sized accordingly).
- 2.4 GHz band initial; external antenna vertical polarization.
- Safety/operations: defined takeoff/landing procedure + keep-away distance from personnel.
- No external infrastructure permitted (closed offline network).

Pre-planning: coverage & recommended UAV count

- **Input:** rough area of operation, expected team positions/routes, and terrain/vegetation constraints (engineering assumptions).
- **Candidate UAV positions:** generate a simple grid of hover points + altitude bands within allowed airspace.
- **Link feasibility (engineering-grade):** use a calibrated path-loss model + safety margin (validated by walk tests / RSSI logs).
- **Output:**
  - recommended initial hover position(s)
  - recommended number of UAV relays (minimum needed to connect the expected teams)
  - spacing rule-of-thumb (Team↔UAV and UAV↔UAV)

Why pre-planning

Reduces time-to-connect in the field; provides a starting point before runtime self-repair takes over.

Deployment workflow (field procedure)

0 — Pre-plan relay placement (before field deployment)

- Load area + constraints
- Compute recommended UAV count + initial hover points

1 — Bring up command and services

- Power command node
- Confirm PTT/map/monitoring status

2 — Deploy ground nodes

- Position team nodes and ground relay
- Confirm baseline routes and voice path

3 — Detect partition event

- Topology table split observed
- Ping + PTT fail between teams

4 — Launch UAV repair

- UAV relay joins mesh and starts autonomous monitoring
- Partition detected → UAV autonomously repositions (direction + altitude band) until thresholds recover
- Route, ping, and PTT recover; operator verifies voice audibility (post-fact evidence)

UAV relay positioning: autonomous self-repair

- **Inputs:** GPS + RSSI/SNR + BATMAN TQ + optional RTT/loss probes from mesh nodes.
- **Partition detection:** missing originators / TQ below threshold / probes fail.
- **Repair loop:**
  - compute link-margin score (maximize the weakest required link)
  - move in small steps (horizontal + altitude band) to improve score
  - stop when routes and PTT are stable for a fixed window
- **Acceptance check:** fail before repair; pass after repair (route + ping + PTT).

Multi-team & multi-UAV operation

- Scaling model: each team has a Team Node; partitions appear when ground links drop below quality thresholds.
- UAV relays act as standard BATMAN peers; multiple UAVs extend coverage by forming an aerial chain (UAV–UAV mesh).
- UAV relays autonomously reposition to reconnect partitions; multiple UAVs can form an aerial chain to extend coverage.
- Telemetry inputs: RSSI/SNR per UAV link, BATMAN TQ + originator visibility, packet loss/RTT probes.
- Deterministic rule: define required links, compute link-margin score per candidate position, move UAV to maximize the weakest required link.
- Spacing emerges from threshold compliance: keep Team↔UAV and UAV↔UAV hops within pass/fail limits shown on the dashboard.
- Acceptance evidence: thresholds met → routes restored → stable end-to-end PTT.

Competitors / previous work and engineering gaps

Named competitors / references

- ResponDrone
- Persistent Systems
- Rajant
- goTenna-like off-grid mesh
- Elsight-like aggregation

Competitor-specific gaps

- ResponDrone: focuses on drone support; lacks open repeatable offline PTT+monitoring kit with acceptance tests.
- Persistent Systems: strong MANET but proprietary/high cost; not reproducible commodity build.
- Rajant: industrial mesh, proprietary ecosystem; not designed for rapid SAR kit deployment with UAV repair proof.
- goTenna-like: messaging-centric; lacks live PTT + topology observability + mesh repair with UAV.
- Elsight-like: connectivity aggregation oriented; not a closed offline network.

Technical challenges and mitigations

A) Software/networking challenges

- Reproducible BATMAN/OpenWrt configs; QoS under loss; offline service discovery.
- Mitigations: scripted configs, bench tests, acceptance harness.

B) Hardware/environment challenges

- Power autonomy vs performance; enclosure/weather/strain relief; antenna placement.
- Mitigations: power logging, simple ruggedization, deployment kit.

C) RF/field challenges

- Vegetation/terrain loss; intermittent links; determining node spacing + UAV hover altitude guidelines.
- Mitigations: walk tests, RSSI/SNR logs, spacing rules.

Partners, stakeholders, and interview outputs

Partner	Requested contribution
ANEPC	Scenario definition, mission constraints, drill realism feedback
GNR/UEPS	Field usability feedback for mountain SAR workflow
Portuguese Air Force	Airspace coordination and UAV operational constraints
Instituto de Telecomunicações	RF guidance, propagation interpretation, measurement support
UAV industry	Payload integration best practices and mounting reliability checks

Requirements

- Partition detection procedure $\leq$ 20 s.
- UAV boot-to-mesh $\leq$ 60 s.
- Reconvergence $\leq$ 30 s after UAV joins.
- Deployment $\leq$ 10 min from transport state.
- Ground autonomy $\geq$ 8 h.
- Payload mass $<$ 500 g.
- Autonomous repair: restore routes/PTT without operator connectivity to the UAV.
- Scalability: supports $\geq$ 2 teams and $\geq$ 1 UAV relay; additional UAV relays integrate as standard mesh relays.

Testing and validation metrics: acceptance matrix

Objective proof	Pass criteria
Pre-UAV partition proof	No route; ping fails; PTT fails; topology table split.
Hard link thresholds validation	RSSI $\geq$ -75 dBm, SNR $\geq$ 20 dB, BATMAN TQ $\geq$ 200/255, probe loss $\leq$ 10%, median RTT $\leq$ 50 ms (30 s window).
UAV entry timing	UAV relay joins mesh in $\leq$ 60 s.
Post-UAV recovery proof	Route restored; ping succeeds; PTT succeeds.
Reconvergence timing	Recovery in $\leq$ 30 s after UAV join.
Deployment timing	Full setup in $\leq$ 10 min.
Autonomy run	Ground nodes operate $\geq$ 8 h with logs.

Primary success criterion: fail before UAV, pass after UAV

Operational KPIs (demonstration layer)

- Time-to-detect partition event (targeted and logged from first route/PTT failure signal).
- Time-to-restore PTT after autonomous UAV repositioning.
- Autonomous repair success rate when operator link is unavailable (repeatable trial count).
- Planning usefulness: time saved in the field using pre-planned initial hover points vs no pre-plan.
- Maximum separation distance where restoration still succeeds under test conditions.
- Setup time from transport state to working closed offline network.
- UAV mission time achieved with payload battery under relay operation.

Operational success demonstration

Before UAV repositioning: PTT across partition fails. After autonomous repositioning: PTT works and operator confirms audibility.

Work plan

- **Phase 1 (Late Feb–Early Mar):** refine proposal, stakeholder interviews, requirement freeze.
- **Phase 2 (Mar):** material procurement and bench setup.
- **Phase 3 (Early Apr):** integration checkpoint and architecture lock.
- **Phase 4 (Late Apr–Mid May):** subsystem development and system integration.
- **Phase 5 (Mid–Late May):** functional prototype tests and defect closure.
- **Phase 6 (Early Jun):** communication assets and final demonstration package.

Course calendar mapping (deliverables)

Milestone date	Concrete output
3-Mar-2026	Revised proposal deck + clarified subsystem boundaries and responsibilities.
10-Mar-2026	Interview blog report with stakeholder summaries and extracted requirement candidates.
17-Mar-2026	Requirements list baseline (functional, operational, and acceptance criteria).
24-Mar-2026	BOM freeze with selected models or documented candidates and procurement plan.
7-Apr-2026	Mid-program pitch final (architecture, validation plan, risks, schedule).
20-Apr-2026	Prototype development start (branch/repo structure, bring-up scripts, bench rigs).
19-May-2026	First functional prototype (offline mesh + PTT + basic telemetry + UAV relay join).
26-May-2026	Prototype + tests done (acceptance matrix + operational KPI datasets).
2-Jun-2026	Communication materials (user workflow docs, visuals, draft video/poster assets).
9-Jun-2026	Final materials package (report, slides, manuals, reproducibility artifacts).
15-Jun-2026	Discussions and final technical review notes integrated.
27-Jun-2026	ElectroDay live demo package and field-ready deployment checklist.

Member responsibilities (1/3)

Pedro Gonçalves — Networking + integration

- Offline mesh bring-up (OpenWrt + BATMAN-adv) reproducible configuration package.
- Partition detection logic + event logging (pre/post UAV evidence).
- End-to-end system integration (network + services + telemetry agents).
- Integration test plan + final integration sign-off checklist.

Henrique Ferreira — Voice + User Workflow Lead

- PTT service deployment (server + clients) packaged for offline operation.
- User onboarding workflow (phone connect, PTT join, troubleshooting steps).
- Voice quality test procedure + dataset (before/after UAV recovery).
- Operator SOP draft → final (deployment steps + safety + pass/fail checks).

Member responsibilities (2/3)

Leonardo Catarino — UAV Payload & Flight Ops Lead

- UAV payload mechanical integration (mount, vibration isolation, antenna placement).
- Payload power system + runtime validation ($\geq$ 30 min target).
- Flight operations checklist + safety procedure (takeoff/landing + keep-away).
- Air-to-ground link characterization logs for autonomy tuning (link thresholds + altitude bands).

Vicente Ferreira — Field Hardware & Deployment Lead

- Ground node ruggedization/enclosure + mast/tripod mounting kit.
- Field deployment kit assembly (cabling, antennas, labels, spares).
- Outdoor setup timing validation ($\leq$ 10 min goal) + reliability notes.
- Mechanical integration documentation (photos, diagrams, build notes).

Member responsibilities (3/3)

Eduardo Curião — Power & Autonomy Lead

- Power budget model for ground nodes + payload (spreadsheet + assumptions).
- Instrumentation harness (voltage/current logging method) for ground + UAV payload.
- Autonomy run tests ($\geq$ 8 h ground) + plots/tables.
- Power impact evaluation of autonomous repair maneuvers (mission time trade-off).
- Battery safety + charging/handling guidelines for team operations.

Dylan Félix — Monitoring + Guidance Software Lead

- Node telemetry agent (GPS + link metrics) and collection pipeline (offline).
- Command dashboard UI (map + status + "PTT OK/FAIL" indicators).
- UAV positioning engine: autonomous partition detection + self-repair repositioning routine.
- Acceptance test harness automation + demo-mode "one-button" bring-up scripts.

Per-member schedule by phase

Member	Late Feb–Early Mar	Mar	Early Apr	Late Apr–Mid May	Mid–Late May	Early Jun
Pedro Gonçalves	Routing baseline doc	Mesh script set v1	Integration test plan v1	Partition pipeline code	Reconvergence evidence logs	Network appendix final
Henrique Ferreira	PTT workflow draft	PTT server config pack	Operator SOP v1	QoS tuning scripts	Voice KPI dataset	User guide final
Eduardo Curião	Power model draft	Instrumentation rig build	Telemetry schema v1	Autonomy run scripts	Battery dataset + plots	Power report final
Vicente Ferreira	Enclosure requirements note	Node kit BOM + CAD	Assembly checklist v1	3-node assembly batch	Robustness test logs	Hardware manual final
Leonardo Catarino	Payload constraints sheet	Mount design prototype	Flight checklist v1	UAV relay integration build	Link characterization logs	UAV ops annex final
Dylan Félix	CI/build workflow draft	Automation scripts v1	Monitoring UI mockups	Integration branch release	Acceptance harness logs	Final package manifest

Conclusion

- We will deliver a deployable, closed offline mesh kit where UAVs are essential for topology repair in partitioned SAR scenarios.
- The implementation focus is operational: pre-plan initial relay placement, detect partitions, autonomously repair connectivity, and verify restored end-to-end PTT audibility.